43.3: Adaptive immunity defends against infection of body fluids and body cells

Having considered how clones of lymphocytes arise, we now explore how these cells help fight infections and minimize damage by pathogens. The defenses provided by B and T lymphocytes can be divided into humoral and cell-mediated immune responses. The **humoral immune response** protects the blood and lymph (once called body *humors*, or fluids). In this response, antibodies help neutralize or eliminate toxins and pathogens in body fluids. In the **cell-mediated immune response**, specialized T cells destroy infected host cells. Both humoral and cellular immunity can include a primary and a secondary immune response, with memory cells enabling the secondary response.

Helper T Cells: Activating Adaptive Immunity

A type of T cell called a **helper T cell** activates humoral and cell-mediated immune responses. Before this can happen, however, two conditions must be met. First, a foreign molecule must be present that can bind specifically to the antigen receptor of the helper T cell. Second, this antigen must be displayed on the surface of an **antigen-presenting cell**. An antigen-presenting cell can be a dendritic cell, macrophage, or B cell.

Like immune cells, infected cells can display foreign antigens on their surface. What then distinguishes antigen-presenting cells from infected cells? The answer lies in two distinct types, or *classes*, of MHC molecules. Most body cells have only the class I MHC molecules, but antigen-presenting cells have both class I and class II MHC molecules. The class II molecules provide a molecular signature by which an antigen-presenting cell is recognized.

A helper T cell and the antigen-presenting cell displaying its specific epitope have a complex interaction. The antigen receptors on the surface of the helper T cell bind to the antigen fragment and to the class II MHC molecule displaying that fragment on the antigen-presenting cell **(Figure 43.18)**. At the same time, an accessory protein called CD4 on the helper T cell surface binds to the class II MHC molecule, helping keep the cells joined. As the two cells interact, signals are exchanged in the form of cytokines. For example, the cytokines secreted from a dendritic cell act in combination with the antigen to stimulate the helper T cell, causing it to produce its own set of cytokines. Also, extensive contact between the cell surfaces enables further information exchange.

Figure 43.18

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The central role of helper T cells in humoral and cell-mediated immune responses.
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Specific helper T cell binds an infected cell, which eventually leads to the proliferation of activated helper T cells that activate humoral and cell-mediated responses.

Here, a helper T cell responds to a dendritic cell displaying an antigen.

The process is as follows.

- An antigen-presenting cell engulfs a pathogen, degrades it, and displays antigen fragments complexed with class II MHC molecules of the cell surface. A specific helper T cell binds to this complex via its antigen receptor and an accessory protein (called CD4).
- Binding of the helper T cell promotes secretion of cytokines by the antigen-presenting cell. These cytokines, along with cytokines from the helper T cell itself, activate the helper T cell and stimulate its proliferation.
- Cell proliferation produces a clone (cluster) of activated helper T cells. All cells in the clone have receptors for the same antigen fragment complex with the same antigen specificity. These cells secrete other cytokines, which help activate B cells (involved in humoral immunity and the secretion of antibodies by plasma cells) and cytotoxic T cells (involved in cell-mediated immunity and an attack on infected cells).

BioFlix® Animation

Activation of a Helper T Cell by a Dendritic Cell (Example: Infection by a Rhinovirus)

image

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image

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Antigen-presenting cells interact with helper T cells in several different contexts. Antigen presentation by a dendritic cell or macrophage activates a helper T cell, which then proliferates, forming a clone of activated cells. In contrast, B cells present antigens to *already* activated helper T cells, which in turn activate the B cells. Activated helper T cells also help stimulate cytotoxic T cells, as we'll discuss shortly.

B Cells and Antibodies: A Response to Extracellular Pathogens

Secretion of antibodies by clonally selected B cells is the hallmark of the humoral immune response. It begins with activation of the B cells.

Activation of B Cells

As illustrated in **Figure 43.19**, activation of B cells involves both helper T cells and proteins on the surface of pathogens. Stimulated by both an antigen and cytokines, the B cell proliferates and differentiates into memory B cells and antibody-secreting plasma cells.

Figure 43.19

Activation of a B cell in the humoral immune response.

Helper T-cell activated with help of cytokines secreted by antigen-presenting cell; activated helper T cell then activates appropriate B cells. Memory B cells and plasma cells proliferate; antibodies secreted.

Most protein antigens require activated helper T cells to trigger a humoral response. A macrophage (shown here) or a dendritic cell can activate a helper T cell, which in turn can activate a B cell to give rise to antibody-secreting plasma cells.

The process is as follows.

- After an antigen-presenting cell (here, a macrophage) engulfs and degrades a pathogen, it displays an antigen fragment complexed with a class II MHC molecule. A helper T cell that recognizes the complex is activated with the aid of cytokines secreted from the antigen-presenting cell.
- When a B cell with receptors for the same epitope internalizes the antigen, it displays an antigen fragment on the cell surface in a complex with a class II MHC molecule. An activated helper T cell bearing receptors specific for the displayed fragment binds to and activates the B cell by secreting cytokines.
- The activated B cell proliferates and differentiates into memory B cells (with Y shaped antigen receptors on surface) and antibody-secreting plasma cells (Y shaped antigen receptors are secreted free of plasma cells as antibodies). The secreted antibodies are specific for the same antigen that initiated the response.

Looking at the steps in this figure, propose a function for the cell-surface antigen receptors of memory B cells.

BioFlix® Animation

Adaptive Defenses: B Cells (Example: Infection by a Rhinovirus)

image

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The pathway for antigen processing and display in B cells differs from that in other antigen-presenting cells. A macrophage or dendritic cell can present fragments from a wide variety of protein antigens, whereas a B cell presents only the antigen to which it specifically binds. When an antigen first binds to receptors on the surface of a B cell, the cell takes in a few foreign molecules by receptor-mediated endocytosis (see Figure 7.19). The class II MHC protein of the B cell then presents an antigen fragment to a helper T cell. This direct cell-to-cell contact is usually critical to B cell activation (see step 2 in Figure 43.19).

B cell activation leads to a robust humoral immune response: A single activated B cell gives rise to thousands of identical plasma cells. These plasma cells stop expressing a membrane-bound antigen receptor and begin producing and secreting antibodies (see step 3 in Figure 43.19). Each plasma cell secretes approximately 2,000 antibodies every second during its four- to five-day life span, nearly a trillion antibody molecules in total. Furthermore, most antigens recognized by B cells contain multiple epitopes. An exposure to a single antigen therefore normally activates a variety of B cells, which give rise to different plasma cells producing antibodies directed against different epitopes on the common antigen.

Antibody Function

Antibodies do not directly kill pathogens, but by binding to antigens, they interfere with pathogen activity or mark pathogens in various ways for inactivation or destruction. Consider, for example, *neutralization*, a process in which antibodies bind to proteins on the surface of a virus **(Figure 43.20a)**. The bound antibodies prevent viral infection of a host cell, thus neutralizing the virus. Similarly, antibodies sometimes bind to toxins released in body fluids, preventing the toxins from entering body cells.

Figure 43.20

Two mechanisms of antibody function.

Two mechanisms of antibody function.

Part a. Neutralization. Antibodies (Y-shaped) bound to antigens on the surface of a virus neutralize it by blocking its ability to bind to a host cell.

Part b. Opsonization. Binding of antibodies to antigens on the surface of bacteria promotes phagocytosis by macrophages and neutrophils.

In *opsonization*, antibodies that are bound to antigens on bacteria do not block infection, but instead present a readily recognized structure for macrophages or neutrophils, thereby promoting phagocytosis **(Figure 43.20b)**. Because each antibody has two antigen-binding sites, antibodies can also facilitate phagocytosis by linking bacterial cells, viruses, or other foreign substances into aggregates.

When antibodies facilitate phagocytosis, as in opsonization, they also help fine-tune the humoral immune response. Recall that phagocytosis enables macrophages and dendritic cells to present antigens to and stimulate helper T cells, which in turn stimulate the very B cells whose antibodies contribute to phagocytosis. This positive feedback between innate and adaptive immunity contributes to a coordinated, effective response to infection.

Antibodies sometimes work together with the proteins of the complement system **(Figure 43.21)**. (The name *complement* reflects the fact that these proteins add to the effectiveness of antibody-directed attacks on bacteria.) Binding of a complement protein to an antigen-antibody complex on a foreign cell triggers events leading to formation of a pore in the membrane of the cell. Ions and water rush into the cell, causing it to swell and lyse.

Figure 43.21

Activation of complement system and pore formation.

Activation of complement system and pore formation.

Binding of antibodies to antigens on the surface of a foreign cell activates the complement system. Here, a complex of complement proteins is bound to antibodies that are bound to antigens in a foreign cell membrane. Some complement proteins are released following activation of the complement system, and form a membrane attack complex in pores in the foreign cell's membrane, allowing water and ions to rush in. The cell swells and eventually lyses.

Although antibodies are the cornerstones of the response in body fluids, there is also a mechanism by which they can bring about the death of infected body cells. When a virus uses a cell's biosynthetic machinery to produce viral proteins, these viral products can appear on the cell surface. If antibodies specific for epitopes on these viral proteins bind to the exposed proteins, the presence of bound antibody at the cell surface can recruit a natural killer cell. The natural killer cell then releases proteins that cause the infected cell to undergo apoptosis. Thus the activities of the innate and adaptive immune systems are once again closely linked.

B cells can express five different types, or *classes*, of immunoglobulin (IgA, IgD, IgE, IgG, and IgM). For a given B cell, each class has an identical antigen-binding specificity but a distinct heavy-chain C region. The B cell antigen receptor, known as IgD, is exclusively membrane bound. The other four classes of Ig have soluble forms, such as the antibodies found in blood, tears, saliva, and breast milk.

Cytotoxic T Cells: A Response to Infected Host Cells

In the absence of an immune response, pathogens can reproduce in and kill infected cells. In the cell-mediated immune response, **cytotoxic T cells** use toxic proteins to kill cells infected by viruses or other intracellular pathogens before pathogens fully mature **(Figure 43.22)**. To become active, cytotoxic T cells require signals from helper T cells and interaction with an antigen-presenting cell. Fragments of foreign proteins produced in infected host cells associate with class I MHC molecules and are displayed on the cell surface, where they can be recognized by activated cytotoxic T cells. Like helper T cells, cytotoxic T cells have an accessory protein that can bind to an MHC molecule. The interaction of this accessory protein, called CD8, with a class I MHC molecule displaying antigen targets the infected cell for cytotoxic T cell activity.

Figure 43.22

The killing action of cytotoxic T cells on an infected host cell.

An activated cytotoxic T cell binds to an infected cell via its antigen receptor and an accessory protein, eventually initiating apoptosis and cell death in the infected cell.

An activated cytotoxic T cell releases molecules that make pores in an infected cell's membrane and enzymes that break down proteins, promoting the cell's death.

The process is as follows.

- An activated cytotoxic T cell binds to a class I MHC-antigen fragment complex on an infected cell via its antigen receptor and an accessory protein called CD8.
- While still bound to the infected cell, the T cell releases perforin molecules, which form pores in the infected cell membrane, and granzymes, enzymes that break down proteins. Granzymes enter infected cells by endocytosis.
- The granzymes initiate apoptosis within the infected cell, leading to fragmentation of the nucleus and cytoplasm and eventual cell death. The released cytotoxic T cell can attack other infected cells.

BioFlix® Animation

Adaptive Defenses: Cytotoxic T Cells (Example: Infection by a Rhinovirus)

image

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The targeted destruction of an infected host cell by a cytotoxic T cell involves the secretion of proteins that disrupt membrane integrity and trigger cell death (apoptosis; see Figure 43.22). The death of the infected cell not only deprives the pathogen of a place to multiply but also exposes cell contents to circulating antibodies, which mark released antigens for disposal.

Summary of the Humoral and Cell-Mediated Immune Responses

As noted earlier, both humoral and cell-mediated immunity can include primary as well as secondary immune responses. Memory cells of each type—helper T cell, B cell, and cytotoxic T cell—enable the secondary response. For example, when body fluids are reinfected by a pathogen encountered previously, memory B cells and memory helper T cells initiate a secondary humoral response. **Figure 43.23** summarizes adaptive immunity, reviews the events that initiate humoral and cell-mediated immune responses, highlights the difference in response to pathogens in body fluids versus in body cells, and emphasizes the central role of the helper T cell.

Figure 43.23

An overview of the adaptive immune response.

Flow diagram of adaptive immune response to an antigen through first and second exposure including both humoral and cell-mediated immune responses.

Flow diagram of adaptive immune response from first and second exposure to an antigen through both humoral and cell-mediated immune responses.

- First exposure to an antigen. Stimulates B cells (humoral response). Gives rise to: Plasma cells (secrete antibodies). Humoral response. Gives rise to: Memory B cells. (Humoral response). Gives rise to antigen being engulfed by an antigen-presenting cell. Stimulates helper T cells. Gives rise to: Memory helper T cells. Memory helper T cells stimulate Memory B cells (humoral response) and memory cytotoxic T cells (cell-mediated response). Stimulates cytotoxic T cells (Cell-mediated response). Gives rise to: Active cytotoxic T cells (cell-mediated response).
- Second exposure to antigen. Simulates memory B cells (humoral response). Gives rise to plasma cells that secrete antibodies; (humoral response). Stimulates memory cytotoxic T cells (cell-

mediated response). Gives rise to active cytotoxic T cells (cell-mediated response). Stimulates memory helper T cells. Gives rise to helper T cells.

VISUAL SKILLS *Identify each arrow as representing part of the primary response or secondary response.*

BioFlix® Animation

Summary of the Adaptive Immune Response (Example: Infection by a Rhinovirus)

image

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Immunization

The protection provided by a second immune response provides the basis for **immunization**, the use of antigens artificially introduced into the body to generate an adaptive immune response and memory cell formation. In 1796, Edward Jenner noted that milkmaids who had cowpox, a mild disease usually seen only in cows, did not contract smallpox, a far more dangerous disease. In the first documented immunization (or *vaccination*, from the Latin *vacca,* cow), Jenner used the cowpox virus to induce adaptive immunity against the closely related smallpox virus. The *vaccines* used today for immunizations may be made from inactivated bacterial toxins, killed or weakened pathogens, or the mRNA or DNA encoding these proteins. COVID-19 vaccines use many of these strategies. For example, the SinoVac and Sinopharm vaccines developed in China use attenuated (weakened) SARS-CoV-2. The most common vaccines in North America specifically target the Spike glycoprotein on the surface of the virus (see Figure 19.2) that enables it to gain access to human cells. Instead of containing the Spike protein itself, these viruses contain the DNA (Janssen/Johnson & Johnson vaccine) or mRNA (Pfizer-BioNTech and Moderna vaccines) that encodes this protein. The recipient's own cells use this genetic material to make Spike proteins, which then stimulate an immune response in the body, as illustrated for an mRNA vaccine in **Figure 43.23X**. This technology has been under development for decades for the treatment of other disorders. Vaccine researchers capitalized on this earlier work to develop safe and effective vaccines relatively rapidly. Since it is much easier to make an mRNA or DNA molecule than a protein molecule in the laboratory, these genetic vaccines can be readily modified to target new viral strains as they emerge. As with most vaccines, maintaining immunity against

SARS-CoV-2 requires the administration of booster vaccine doses. These boosters trigger a secondary immune response, replenishing stores of memory cells (see Figure 43.17).

Figure 43.23X

mRNA vaccines stimulate both innate and adaptive immune responses.

The figure illustrates the stimulation of innate and adaptive immune responses by m R N A vaccines.

COVID-19 mRNA vaccines consist of an mRNA molecule encoding the viral Spike (S) protein contained within a tiny lipid sphere.

Step 1: When a dendritic cell takes up the lipid sphere, the m R N A is released into the cytosol. Numerous small spherical lipid spheres containing S-protein m R N A enter inside a dendritic cell, which has multiple cytoplasmic projections.

Step 2: Cytosolic Toll-like receptors (T L R s) recognize the m R N A molecule as foreign, and stimulate the production of interferons and other cytokines of the innate immune response. These chemicals stimulate the proliferation and activity of all types of lymphocytes. They also stimulate the inflammatory responses that are experienced as side effects, such as shoulder pain, fever, and general malaise. A single helical strand of S-protein m R N A is bound to a T L R receptor. Several small molecules labeled cytokines are released from the dendritic cell upon activation of the T L R.

Step 3: Dendritic cell ribosomes translate the m R N A into S-proteins, some of which are inserted into the plasma membrane. A single helical S-protein m R N A strand is bound to the subunits of the ribosome. A small chain of polypeptides constituting the S protein is formed.

Step 4: S-proteins bind to B cells with the matching receptor. A large nucleated B cell has two Y-shaped surface projections depicting receptors. An S-protein binds to one receptor.

Step 5: These B cells proliferate and differentiate into plasma cells that produce antibodies that recognize S proteins or into memory B cells. (Note that full activation of the B cells also requires binding by S-protein-specific helper T cells, a process not shown here.) B cells bound to S-protein form memory B cells and plasma cells. The plasma cells produce antibodies that recognize the S-protein.

Step 6: Other S-proteins are broken down into smaller fragments and presented on the cell surface bound to M H C class Roman numeral 2 receptors, which bind T cells with the matching receptor. Small degraded fragments of an S-protein are formed. A nucleated T cell has two surface receptors. An S-protein fragment is bound to one receptor on the T cell.

Step 7: These activated T cells proliferate and differentiate into helper T cells, cytotoxic T cells, and memory cells. A second (booster) shot activates the memory B and T cells, resulting in the rapid production of all lymphocyte types, including more memory cells. As a result, the booster provides a more robust and long-lasting immunity. The activated T cell forms numerous memory T cells, cytotoxic T cells, and helper T cells. Helper T-cells release cytokines.

Vaccination programs have been successful against many infectious diseases that once killed or incapacitated large numbers of people. A worldwide vaccination campaign led to eradication of smallpox in the late 1970s. In industrialized nations, routine immunization of infants and children has dramatically reduced the incidence of sometimes devastating diseases, such as polio and measles **(Figure 43.24)**. Combined with public health measures such as masks and physical distancing, the COVID-19 vaccine campaign has been an important measure in fighting the pandemic. However, this massive effort was hampered by vaccine hesitancy and by global inequities in vaccine distribution. Ensuring global COVID-19 vaccine availability is not only a humanitarian initiative. The more infections, the higher the likelihood that concerning variants will develop, as we will discuss shortly. Therefore, widespread vaccination is crucial to prevent the rise and spread of viral variants that could evade existing vaccines and thus spark new outbreaks.

Figure 43.24

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    Vaccine-based protection against two life-threatening
communicable diseases.
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Two bar charts plot annual deaths from polio and measles in the United States by year and corresponding world maps are shown.

The graphs show deaths by year in the United States caused by polio and measles. The maps show examples of the global progress against these two diseases.

Annual deaths from polio were between 1000 and 3200 in the 1950s. A vaccine was introduced in 1956 and deaths dropped below 500, continually tapering off until the disease was eradicated in the United States in 1988. By that time, much of Europe, Australia, Japan, and Argentina had also eradicated it. By 2014, all by 3 countries eliminated the disease. Annual deaths from measles in the United States bounced between 350 and 700 from 1950 through 1965. A vaccine was introduced around 1964. Deaths tumbled with small spikes in 1970, 1971, 1989, 1990, and 1991. By 2014, much of the Western world, Russia, China, South America, and northern Africa had a 90 to 100% measles vaccination rate. India, some South American countries, and

some African countries were between 80 and 89%. Some eastern European countries, several African countries, and a few Asian countries were between 50 and 79%, with several in Africa falling below 50%.

Vaccine hesitancy is fueled by misinformation about safety and disease risk. Consider measles as just one example. The vaccine is highly effective and very safe—fewer than one in a million children experience a significant allergic reaction. Indeed, a vaccination campaign had effectively eliminated the disease in the United States by 2000. However, the disease remains quite dangerous to this day, killing more than 200,000 people worldwide each year. Recently, a decline in vaccination rates in the United States has led to multiple outbreaks since 2010. A multi-state outbreak in 2019 was the worst in two decades, with more than 1,100 cases in the first six months of the year. Even with modern medical care, measles kills on average one out of every 1,000 infected individuals.

Active and Passive Immunity

Our discussion of adaptive immunity has to this point focused on **active immunity**, the defenses that arise when a pathogen infection or immunization prompts an immune response. A different type of immunity results when, for example, the IgG antibodies in the blood of a pregnant female cross the placenta to her fetus. This protection is called **passive immunity** because the antibodies in the recipient (in this case, the fetus) are produced by another individual (the mother). IgA antibodies present in breast milk provide additional passive immunity to the infant's digestive tract while the infant's immune system develops. Passive immunity persists only as long as the transferred antibodies last (up to a few months).

In artificial passive immunization, antibodies from an immune animal are injected into a nonimmune animal. For example, humans bitten by venomous snakes are sometimes treated with antivenin, serum from sheep or horses that have been immunized against a snake venom. When injected immediately after a snakebite, the antibodies in antivenin can neutralize toxins in the venom before the toxins do massive damage.

Antibodies as Tools

Antibodies that an animal produces after exposure to an antigen are the products of many different clones of plasma cells, each specific for a different epitope. However, antibodies can also be prepared from a single clone of B cells grown in culture. The **monoclonal antibodies** produced by such a culture are identical and specific for the same epitope on an antigen.

Monoclonal antibodies have provided the basis for many recent advances in medical diagnosis and treatment. For example, home pregnancy test kits use monoclonal antibodies to detect human chorionic gonadotropin (hCG). Because hCG is produced as soon as an embryo implants in the uterus (see Concept

46.5), the presence of this hormone in a woman's urine is a reliable indicator for a very early stage of pregnancy. Monoclonal antibodies are also being used as therapies for a growing number of human diseases, including many cancers and COVID-19.

One recently developed antibody tool uses a single drop of blood to identify every virus that a person has encountered through infection or vaccination **(Figure 43.25)**. To detect the antibodies formed against these viruses, researchers generated a set of nearly 100,000 bacteriophages, each of which displays a different peptide from one of the roughly 200 species of viruses that infect humans.

Figure 43.25

A comprehensive test for past viral encounters.

A comprehensive test for past viral encounters.

By combining the power of DNA sequencing with the specificity of antigen recognition by antibodies, researchers can identify every virus that a person's immune system has encountered during the person's lifetime.

The process is as follows.

- Viruses that infect humans have unique peptides on their surface. The virus on the left is shown as a yellow spiky circle; the virus in the middle is a blue rod; and the virus on the right is a green oval. By introducing short DNA sequences from all known human viruses into copies of a bacteriophage genome, researchers generated a collection of 100,000 bacteriophages, each displaying many copies of one viral peptide.
- The bacteriophages are combined with serum from a drop of a person's blood. The serum contains antibodies, some of which were produced in response to exposure to viruses. Any antibody that is specific for a viral peptide binds to a bacteriophage displaying that peptide. Bacteriophages displaying peptides from viruses never encountered are not recognized.
- DNA sequencing of bacteriophages to which antibodies are bound identifies the complete set of viruses to which a person has been exposed. In this case, this person has been exposed to two viruses, the yellow spiky circle and the green oval, but not to the blue rod.

WHAT IF? *All of the antibodies are shown with just one antigen-binding site occupied. If a single antibody bound to two bacteriophages, how would this affect the results?*

*For suggested answer, see*Appendix A*.*

Immune Rejection

Like pathogens, cells from another person can be recognized as foreign and attacked by immune defenses. For example, skin transplanted from one person to a genetically nonidentical person will look healthy for a week or so but will then be destroyed (rejected) by the recipient's immune response. It turns out that MHC molecules are a primary cause of rejection. Why? Each of us expresses MHC proteins from more than a dozen different MHC genes. Furthermore, there are more than 100 different versions, or alleles, of human MHC genes. As a consequence, the sets of MHC proteins on cell surfaces are likely to differ between any two people, except identical twins. Such differences can stimulate an immune response in the recipient of a transplant or graft, causing rejection. To minimize rejection of a transplant or graft, surgeons use donor tissue bearing MHC molecules that match those of the recipient as closely as possible. In addition, the recipient takes medicines that suppress immune responses (but as a result leave the recipient more susceptible to infections).

Blood Groups

In the case of blood transfusions, the recipient's immune system can recognize glycoproteins on the surface of blood cells as foreign, triggering an immediate and devastating reaction. To avoid this danger, the so-called ABO blood groups of the donor and recipient must be taken into account. Red blood cells are designated as type A if they have the A carbohydrate on their surface. Similarly, the B carbohydrate is found on the surface of type B red blood cells; both A and B carbohydrates are found on type AB red blood cells; and neither carbohydrate is found on type O red blood cells (see Figure 14.11).

Why does the immune system recognize particular sugars on red blood cells? It turns out that we are frequently exposed to certain bacteria that have epitopes very similar to the carbohydrates on blood cells. A person with type A blood will respond to the bacterial epitope similar to the B carbohydrate and make antibodies that will react with any B carbohydrate encountered upon a transfusion. However, that same person doesn't make antibodies against the bacterial epitope similar to the A carbohydrate because lymphocytes that would be reactive with the body's own cells and molecules were inactivated or eliminated during development.

To understand how ABO blood groups affect transfusions, let's consider further the example of a person with type A blood receiving a transfusion of type B blood. The person's anti-B antibodies would cause the transfused red blood cells to undergo lysis, triggering chills, fever, shock, and perhaps kidney malfunction. At the same time, anti-A antibodies in the donated type B blood would act against the recipient's red blood cells. Applying the same logic to a type O person, we can see that such interactions would cause a problem upon transfusion of any other blood type. Fortunately, the discovery of enzymes that can cleave the A and B carbohydrates from red blood cells may eliminate this problem in the future.